NAME : Syeda Areej E fatima

ROLL NO : 25p0014

Section : BAI-2A

Lab Task : 6

Question 1:

Header file:

class_header.h | main.cpp .../Q1 | class_header.cpp | book_

lab_tasks > labtask6 > Q1 > class_header.h > ...

```cpp
#include<iostream>
#include<string>
using namespace std;

class Student
{
    private :
        string name;
        int roll;
        int marks;

    public :
        void input();
        void display();
};
```

Cpp file:

class_header.h | main.cpp .../Q1 | class_header.cpp | book_header.h

lab_tasks > labtask6 > Q1 > class_header.cpp > ...

```cpp
#include<iostream>
#include<string>
#include"class_header.h"

using namespace std;

//member function definition

void Student::input()
{
    cout<<"-----Taking Input------"<<endl;
    cout<<"Enter name : ";
    cin>>name;
    cout<<"Enter Roll no. : ";
    cin>>roll;
    cout<<"Enter Marks :";
    cin>>marks;
};

void Student::display()
{
    cout<<"-----Displaying------"<<endl;
    cout<<"name : "<<name<<endl;
    cout<<"Roll no. : "<<roll<<endl;
    cout<<"Marks"<<marks<<endl;
};

```

Main cpp file :

class_header.h | main.cpp .../Q1 | class_

lab_tasks > labtask6 > Q1 > main.cpp > main()

```cpp
#include<iostream>
#include"class_header.h"

using namespace std;

int main()
{
    Student s1;
    s1.input();
    s1.display();

    return 0;
}
```

Output:

```
syeda-areej-e-fatima@velanora:~/Sem_
^[[Asyeda-areej-e-fatima@velanora:~/
-----Taking Input------
Enter name : Waleeja
Enter Roll no. : 5555
Enter Marks :45
-----Displaying------
name : Waleeja
Roll no. : 5555
Marks : 45
```

Question 2:

Header file:

book_header.h × | book.cpp | main.cpp .../Q2 | rectangle_hea

lab_tasks > labtask6 > Q2 > book_header.h > ...

```
#include<iostream>
#include <string>
using namespace std;

class Book
{
    private:
        string title;
        string name;
        int price;

    public :
        void input();
        void display();
};
```

Cpp file:

book_header.h | book.cpp | main.cpp .../Q2 | rectangle_header.h

lab_tasks > labtask6 > Q2 > book.cpp > input()

```cpp
#include<iostream>
#include<limits>
#include<string>
#include"book_header.h"
using namespace std;

void Book::input()
{
    cout<<"-----Taking Input------"<<endl;
    cout<<"Enter title name : ";
    cin.ignore(numeric_limits<streamsize>::max(), '\n');
    getline(cin, title);
    cout<<"Enter author name : ";
    getline(cin, name);
    cout<<"Enter price :";
    cin>>price;
};

void Book::display()
{

    cout << "-----Displaying------" << endl;
    cout << "Title name : " << title << endl;
    cout << "Author name : " << name << endl;
    cout << "Price :" << price << endl;
};
```

Main cpp file :

book_header.h | book.cpp | main.cpp .../Q2 | rectangle_header.h

lab_tasks > labtask6 > Q2 > main.cpp > main()

```cpp
#include<iostream>
#include<string>
#include"book_header.h"

using namespace std;

int main()
{
    Book books[3];

    for (int i = 0; i < 3; i++)
    {
        cout<<"===== BOOK "<<i+1<<" ==== "<<endl;
        books[i].input();
        books[i].display();
        cout<<endl;
    }

    return 0;

}
```

Output:

```
syeda-areej-e-fatima@velanora:~/Sem_2/OOPs_lab/lab_tasks/labtask6/
syeda-areej-e-fatima@velanora:~/Sem_2/OOPs_lab/lab_tasks/labtask6/
===== BOOK 1 ====
-----Taking Input------
Enter title name : Lalalala

Enter author name : Syeda Easha
Enter price :2300
-----Displaying------
Title name :
Author name : Syeda Easha
Price :2300

===== BOOK 2 ====
-----Taking Input------
Enter title name : Nanhay Doggy
Enter author name : Hasnain
Enter price :3200
-----Displaying------
Title name : Nanhay Doggy
Author name : Hasnain
Price :3200

===== BOOK 3 ====
-----Taking Input------
Enter title name : Mai toh abdul bari hoon
Enter author name : Rizvi sahab
Enter price :4400
-----Displaying------
Title name : Mai toh abdul bari hoon
Author name : Rizvi sahab
Price :4400
```

Question 3:

Header file:

rectangle_header.h × | rectangle.cpp | main.cpp .../Q3

lab_tasks > labtask6 > Q3 > rectangle_header.h > ...

```cpp
#include<iostream>

using namespace std;

class Rectangle
{

    private :
        int length;
        int width;

    public :
        void input();
        int calculate_area();

};
```

Cpp file:

rectangle_header.h | rectangle.cpp | main.cpp .../Q3 | resu

lab_tasks > labtask6 > Q3 > rectangle.cpp > calculate_area()

```cpp
#include<iostream>
#include"rectangle_header.h"

using namespace std;

void Rectangle::input()
{
    int l,w;
    cout<<"Enter the length : ";
    cin>>l;
    cout<<"Enter the width : ";
    cin>>w;

    length = l;
    width = w;
}
int Rectangle::calculate_area()
{
   int area = length*width;

   return area;

}
```

Main cpp file:

rectangle_header.h | rectangle.cpp | main.cpp .../Q3 | resu

lab_tasks > labtask6 > Q3 > main.cpp > ...

```cpp
#include<iostream>
#include"rectangle_header.h"
using namespace std;

int main()
{
    Rectangle r1;
    r1.input();
    int area = r1.calculate_area();

    cout<<"Area of rectangle "<<area<<endl;

    return 0;
}
```

Output:

```
syeda-areej-e-fatima@velanora:~/
Enter the length : 6
Enter the width : 4
Area of rectangle 24
syeda-areej-e-fatima@velanora:~/
```

Question 4:

Header file :

result_header.h × result.cpp main.cpp .../Q4_result_management_

lab_tasks > labtask6 > Q4_result_management_system > result_header.h > ...

```cpp
#include<iostream>
#include <string>
using namespace std;

class Result
{
    private :
        string name ;
        int roll_no;
        int subject_marks[5];

    public :
        void input_data();
        float calculate_average();
        void display_result();

};
```

Cpp file :

result_header.h | result.cpp | main.cpp .../ Q4_result_management_system | main.cpp .../Q5_employee_salary_system

lab_tasks > labtask6 > Q4_result_management_system > result.cpp > calculate_average()

```cpp
#include<iostream>
#include"result_header.h"

void Result::input_data()
{
    cout<<"Enter the name : ";
    cin>>name;
    cout<<"Enter the Roll number : ";
    cin>>roll_no;
    for (int i = 0; i < 5; i++)
    {
        cout<<"enter marks of subject "<<i+1<<" : ";
        cin>>subject_marks[i];
        if ( subject_marks[i] <0 || subject_marks[i] >100 )
        {
            cout<<"Invalid marks /n Enter marks of subject "<<i+1<<" again : ";
            cin>>subject_marks[i];
        }
    }
}

float Result::calculate_average()
{
    float avg=1.0;
    float sum=0.0;

    for (int i = 0; i < 5; i++)
    {
        sum += subject_marks[i];

    }
```

```cpp
    }

    avg = sum / (float) 5.0;

    return avg;
}

void Result::display_result()
{                                   // input
    cout<<"----- Marks ----- "<<endl;
    for (int i = 0; i < 5; i++)
    {
        cout<<"Subject "<<i+1<<" : "<<subject_marks[i]<<endl;
    }

    float avg;
    avg = calculate_average();                  // average

    cout<<"Average : "<<avg<<endl;


    if(avg >= 50.0)
    {
        cout<<" PASS "<<endl;
    }
    else
    {
        cout<<" FAIL "<<endl;
    }

}
```

Main Cpp file:

result_header.h | result.cpp | main.cpp .../ Q4_result_management_system | main.c

lab_tasks > labtask6 > Q4_result_management_system > main.cpp > main()

```cpp
#include<iostream>
#include"result_header.h"
using namespace std;

int main()
{
    Result r[2];

    for (int i = 0; i < 2; i++)
    {
        cout<<"======= student "<<i+1<<" ========"<<endl;
        r[i].input_data();
        r[i].display_result();
    }

}
```

Output:

```
syeda-areej-e-fatima@velanora:~/Sem_2/OOPs_lab/lab_tas
======= student 1 ========
Enter the name : anoosha
Enter the Roll number : 3678
enter marks of subject 1 : 34
enter marks of subject 2 : 56
enter marks of subject 3 : 78
enter marks of subject 4 : 46
enter marks of subject 5 : 98
----- Marks -----
Subject 1 : 34
Subject 2 : 56
Subject 3 : 78
Subject 4 : 46
Subject 5 : 98
Average : 62.4
 PASS
======= student 2 ========
Enter the name : sahar
Enter the Roll number : 6754
enter marks of subject 1 : 98
enter marks of subject 2 : 87
enter marks of subject 3 : 98
enter marks of subject 4 : 23
enter marks of subject 5 : 54
----- Marks -----
Subject 1 : 98
Subject 2 : 87
Subject 3 : 98
Subject 4 : 23
Subject 5 : 54
Average : 72
 PASS
syeda-areej-e-fatima@velanora:~/Sem_2/OOPs_lab/lab_tas
```

Question 5:

Header file :

employee.cpp | main.cpp | employee_header.h

lab_tasks > labtask6 > Q5_employee_salary_system > employee_header.h > ..

```cpp
#include<iostream>
#include<string>
using namespace std;


class Employee
{
    private :
        string name;
        int ID;
        int basic_salary;

    public :
        void input_Data();
        int calculate_Bonus();
        void display_Details();


};
```

Cpp file :

employee.cpp × main.cpp employee_header.h

lab_tasks > labtask6 > Q5_employee_salary_system > employee.cpp > input_Data()

```cpp
#include<iostream>
#include<limits>
#include"employee_header.h"
using namespace std;

void Employee::input_Data()
{
    cout<<"Enter name : ";
    cin.ignore(numeric_limits<streamsize>::max(), '\n');
    getline(cin,name);
    cout<<"Enter ID : ";
    cin>>ID;
    cout<<"Enter salary : ";
    cin>>basic_salary;

}

int Employee::calculate_Bonus()
{
    float bonus = basic_salary*10/100;
    return bonus;

}

void Employee::display_Details()
{
    cout<<"Name : "<<name<<endl;
    cout<<"ID : "<<ID<<endl;
    cout<<"basic Salary : "<<basic_salary<<endl;
    int bonus = calculate_Bonus();
    cout<<"Bonus : "<<bonus<<endl;
    int final_salary = basic_salary + bonus;

    cout<<"Final salary : "<<final_salary<<endl;

}
```

Main Cpp file:

```cpp
#include<iostream>
#include"employee_header.h"
using namespace std;

int main()
{
    int N;
    cout<<"Enter number of employees : ";
    cin>>N;

    Employee emp[N];


    for (int i = 0; i < N; i++)
    {
        cout<<"======= Employee "<<i+1<<" ========"<<endl;
        emp[i].input_Data();
        emp[i].display_Details();
    }

```

Output:

```
Enter number of employees : 3
======= Employee 1 ========
Enter name : Yaseen Khan
Enter ID : 15
Enter salary : 50000
Name : Yaseen Khan
ID : 15
basic Salary : 50000
Bonus : 5000
Final salary : 55000
======= Employee 2 ========
Enter name : Waleeja
Enter ID : 5555
Enter salary : 45000
Name : Waleeja
ID : 5555
basic Salary : 45000
Bonus : 4500
Final salary : 49500
======= Employee 3 ========
Enter name : Dua
Enter ID : 3445
Enter salary : 55000
Name : Dua
ID : 3445
basic Salary : 55000
Bonus : 5500
Final salary : 60500
```